

A COUNTEREXAMPLE TO A CONJECTURE OF PETR, PORTIER AND STOLARCZYK

ERIC LEE

For a pebbling distribution P on a graph, the *value* of a vertex x is $v(x) = \sum_y P(y) 2^{-d(x,y)}$. In [1] the following is conjectured.

Conjecture 4.1 of [1]. *Let P be a solvable pebbling distribution on the grid. Then for all vertices x that have degree 4, $v(x) \geq \frac{3}{2}$.*

Theorem 1. *The distribution P in Figure 1 on the 10×10 grid has $|P| = 57$, is solvable, and satisfies $v(5, 5) = \frac{379}{256} < \frac{3}{2}$.*

	0	1	2	3	4	5	6	7	8	9
0	2			2			3		4	
1	3	2	2	3						
2	3	2	2	3						
3	2								4	
4							1			
5						<i>x</i>				
6	1			1				2		
7							1			
8	4			4			1		3	
9									2	

FIGURE 1. The distribution P , with 57 pebbles. Rows and columns are indexed from 0 as shown; empty cells carry no pebble. The shaded vertex $x = (5, 5)$ has degree 4 and carries no pebble.

The value. Summing the contributions $P(y) 2^{-d(x,y)}$ of the 24 occupied vertices y , in row order:

$$\begin{aligned}
 v(5, 5) &= \frac{1}{512} + \frac{1}{64} + \frac{3}{64} + \frac{1}{64} + \frac{3}{512} + \frac{1}{128} + \frac{1}{64} + \frac{3}{64} + \frac{3}{256} + \frac{1}{64} + \frac{1}{32} + \frac{3}{32} \\
 &\quad + \frac{1}{64} + \frac{1}{8} + \frac{1}{4} + \frac{1}{64} + \frac{1}{8} + \frac{1}{4} + \frac{1}{8} + \frac{1}{64} + \frac{1}{8} + \frac{1}{16} + \frac{3}{64} + \frac{1}{64} = \frac{379}{256} \approx 1.4805.
 \end{aligned}$$

It is the only vertex of degree 4 with value below $\frac{3}{2}$.

Solvability. The appendix gives, for each of the 100 vertices, a sequence of moves from P that places a pebble on it (466 moves in total, at most 46 for one vertex). A move $\mathbf{ab} \rightarrow \mathbf{cd}$ removes two pebbles from (a, b) and adds one to (c, d) , and “(has a pebble)” marks vertices that already carry one. Each sequence can be checked by hand; we have also replayed all of them with three independent programs.

Larger grids. We found solvable distributions with a degree-4 vertex of value below $\frac{3}{2}$ on $n \times n$ grids for $n = 14, 18, 24, 32, 40, 48$, with the smallest such value settling near 1.4477, a deficit of about 0.052, from $n = 24$ on. These were certified by two independent solvers. In every example the low vertex lies at distance 6 from the boundary, so the phenomenon appears to be tied to the boundary; we do not know whether such vertices exist far from the boundary. Under the hemmed distribution P' of [1], every degree-4 vertex has value at least 1.97 in the 10×10 to 12×12 examples and about 1.64 in the larger ones.

How it was found. By a randomized local search over solvable distributions on small grids, using moves that compress, split, merge and shift piles of pebbles, with every candidate's solvability decided exactly by two independent solvers.

REFERENCES

- [1] J. Petr, J. Portier and S. Stolarczyk, *A new lower bound on the optimal pebbling number of the grid*, Discrete Math. **346** (2023), 113212.

APPENDIX: A MOVE SEQUENCE FOR EVERY VERTEX

00: (*has a pebble*)
 01: $11 \rightarrow 01$
 02: $12 \rightarrow 02$
 03: (*has a pebble*)
 04: $03 \rightarrow 04$
 05: $06 \rightarrow 05$
 06: (*has a pebble*)
 07: $08 \rightarrow 07$
 08: (*has a pebble*)
 09: $08 \rightarrow 09$
 10: (*has a pebble*)
 11: (*has a pebble*)
 12: (*has a pebble*)
 13: (*has a pebble*)
 14: $13 \rightarrow 14$
 15: $23 \rightarrow 13$ $13 \rightarrow 14$ $13 \rightarrow 14$ $14 \rightarrow 15$
 16: $06 \rightarrow 16$
 17: $06 \rightarrow 07$ $08 \rightarrow 07$ $07 \rightarrow 17$
 18: $08 \rightarrow 18$
 19: $08 \rightarrow 18$ $08 \rightarrow 18$ $18 \rightarrow 19$
 20: (*has a pebble*)
 21: (*has a pebble*)
 22: (*has a pebble*)
 23: (*has a pebble*)
 24: $23 \rightarrow 24$
 25: $13 \rightarrow 23$ $23 \rightarrow 24$ $23 \rightarrow 24$ $24 \rightarrow 25$
 26: $08 \rightarrow 07$ $08 \rightarrow 07$ $07 \rightarrow 06$ $06 \rightarrow 16$ $06 \rightarrow 16$ $16 \rightarrow 26$
 27: $38 \rightarrow 28$ $38 \rightarrow 28$ $28 \rightarrow 27$
 28: $38 \rightarrow 28$
 29: $38 \rightarrow 28$ $38 \rightarrow 28$ $28 \rightarrow 29$
 30: (*has a pebble*)
 31: $21 \rightarrow 31$
 32: $22 \rightarrow 32$
 33: $23 \rightarrow 33$
 34: $13 \rightarrow 23$ $23 \rightarrow 33$ $23 \rightarrow 33$ $33 \rightarrow 34$
 35: $12 \rightarrow 13$ $22 \rightarrow 23$ $08 \rightarrow 07$ $08 \rightarrow 07$ $07 \rightarrow 06$ $06 \rightarrow 05$ $06 \rightarrow 05$ $13 \rightarrow 14$ $13 \rightarrow 14$ $05 \rightarrow 15$ $14 \rightarrow 15$ $23 \rightarrow 24$ $23 \rightarrow 24$
 15 $\rightarrow 25$ $24 \rightarrow 25$ $25 \rightarrow 35$
 36: $00 \rightarrow 10$ $10 \rightarrow 11$ $10 \rightarrow 11$ $30 \rightarrow 20$ $11 \rightarrow 12$ $11 \rightarrow 12$ $20 \rightarrow 21$ $20 \rightarrow 21$ $12 \rightarrow 13$ $03 \rightarrow 13$ $12 \rightarrow 13$ $21 \rightarrow 22$ $21 \rightarrow 22$
 13 $\rightarrow 23$ $22 \rightarrow 23$ $22 \rightarrow 23$ $13 \rightarrow 14$ $13 \rightarrow 14$ $23 \rightarrow 24$ $14 \rightarrow 24$ $23 \rightarrow 24$ $23 \rightarrow 24$ $08 \rightarrow 07$ $08 \rightarrow 07$ $07 \rightarrow 06$ $24 \rightarrow 25$ $24 \rightarrow 25$
 06 $\rightarrow 16$ $06 \rightarrow 16$ $25 \rightarrow 26$ $16 \rightarrow 26$ $26 \rightarrow 36$
 37: $38 \rightarrow 37$
 38: (*has a pebble*)
 39: $38 \rightarrow 39$
 40: $30 \rightarrow 40$
 41: $30 \rightarrow 31$ $21 \rightarrow 31$ $31 \rightarrow 41$
 42: $12 \rightarrow 22$ $23 \rightarrow 22$ $22 \rightarrow 32$ $22 \rightarrow 32$ $32 \rightarrow 42$
 43: $13 \rightarrow 23$ $23 \rightarrow 33$ $23 \rightarrow 33$ $33 \rightarrow 43$

44: 11→21 20→21 21→31 30→31 03→13 12→22 21→22 22→32 31→32 13→23 22→23 13→23 23→33
 32→33 23→33 23→33 33→43 33→43 43→44
 45: 00→10 10→11 10→11 30→20 11→12 11→12 20→21 20→21 12→13 03→13 12→13 21→22 21→22
 13→23 22→23 22→23 13→14 13→14 23→24 14→24 23→24 23→24 08→07 08→07 07→06 24→25 24→25
 06→16 06→16 25→26 16→26 38→37 38→37 26→36 37→36 36→46 46→45
 46: (*has a pebble*)
 47: 00→10 10→11 10→11 30→20 11→12 11→12 20→21 20→21 12→13 03→13 12→13 21→22 21→22
 13→23 22→23 22→23 13→14 13→14 23→24 14→24 23→24 23→24 08→07 08→07 07→06 24→25 24→25
 06→16 06→16 25→26 16→26 38→37 38→37 26→36 37→36 36→46 46→47
 48: 38→48
 49: 38→48 38→48 48→49
 50: 80→70 80→70 70→60 60→50
 51: 11→21 22→21 10→20 21→31 21→31 20→30 20→30 30→40 30→40 40→41 31→41 41→51
 52: 03→13 13→12 13→12 00→10 12→11 10→11 12→22 23→22 10→20 11→21 11→21 20→30 20→30
 21→31 21→31 30→40 30→40 40→41 31→41 22→32 22→32 41→42 32→42 42→52
 53: 83→73 83→73 73→63 63→53
 54: 11→21 20→21 21→31 30→31 03→13 12→22 21→22 22→32 31→32 13→23 22→23 13→23 23→33
 32→33 23→33 23→33 33→43 33→43 83→73 83→73 73→63 43→53 63→53 53→54
 55: 00→10 10→11 10→11 30→20 11→12 11→12 20→21 20→21 12→13 03→13 12→13 21→22 21→22
 13→23 22→23 22→23 13→14 13→14 23→24 14→24 23→24 23→24 08→07 08→07 07→06 24→25 24→25
 06→16 06→16 25→26 16→26 38→37 38→37 26→36 37→36 98→88 88→87 88→87 87→86 86→76 76→66
 67→66 36→46 66→56 46→56 56→55
 56: 00→10 10→11 10→11 30→20 11→12 11→12 20→21 20→21 12→13 03→13 12→13 21→22 21→22
 13→23 22→23 22→23 13→14 13→14 23→24 14→24 23→24 23→24 08→07 08→07 07→06 24→25 24→25
 06→16 06→16 25→26 16→26 38→37 38→37 26→36 37→36 36→46 46→56
 57: 67→57
 58: 38→48 38→48 48→58
 59: 98→88 88→78 88→78 78→68 67→68 38→48 38→48 68→58 48→58 58→59
 60: (*has a pebble*)
 61: 80→70 80→70 70→60 60→61
 62: 83→73 83→73 73→63 63→62
 63: (*has a pebble*)
 64: 83→73 83→73 73→63 63→64
 65: 98→88 88→87 88→87 87→86 86→76 76→66 67→66 66→65
 66: 67→66
 67: (*has a pebble*)
 68: 67→68
 69: 98→88 88→78 88→78 78→68 67→68 68→69
 70: 80→70
 71: 80→81 80→81 81→71
 72: 83→82 83→82 82→72
 73: 83→73
 74: 83→73 83→73 73→74
 75: 98→88 88→87 88→87 87→86 86→76 76→75
 76: (*has a pebble*)
 77: 67→77
 78: 88→78
 79: 98→88 88→78 88→78 78→79
 80: (*has a pebble*)
 81: 80→81

82: $83 \rightarrow 82$
 83: *(has a pebble)*
 84: $83 \rightarrow 84$
 85: $98 \rightarrow 88$ $88 \rightarrow 78$ $88 \rightarrow 78$ $78 \rightarrow 77$ $67 \rightarrow 77$ $77 \rightarrow 76$ $76 \rightarrow 86$ $86 \rightarrow 85$
 86: *(has a pebble)*
 87: $88 \rightarrow 87$
 88: *(has a pebble)*
 89: $88 \rightarrow 89$
 90: $80 \rightarrow 90$
 91: $80 \rightarrow 81$ $80 \rightarrow 81$ $81 \rightarrow 91$
 92: $83 \rightarrow 82$ $83 \rightarrow 82$ $82 \rightarrow 92$
 93: $83 \rightarrow 93$
 94: $83 \rightarrow 84$ $83 \rightarrow 84$ $84 \rightarrow 94$
 95: $98 \rightarrow 88$ $88 \rightarrow 78$ $88 \rightarrow 78$ $78 \rightarrow 77$ $67 \rightarrow 77$ $77 \rightarrow 76$ $76 \rightarrow 86$ $83 \rightarrow 84$ $83 \rightarrow 84$ $84 \rightarrow 85$ $86 \rightarrow 85$ $85 \rightarrow 95$
 96: $98 \rightarrow 88$ $88 \rightarrow 78$ $88 \rightarrow 78$ $78 \rightarrow 77$ $67 \rightarrow 77$ $77 \rightarrow 76$ $76 \rightarrow 86$ $86 \rightarrow 96$
 97: $98 \rightarrow 97$
 98: *(has a pebble)*
 99: $98 \rightarrow 99$